

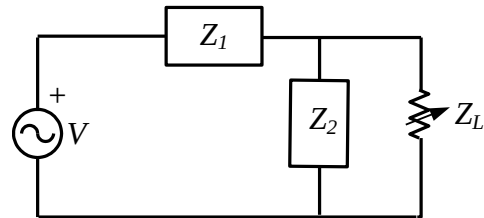
Name:

R.

Section:

1. In the figure Z_L consists of a variable resistance and a fixed capacitive reactance of 8Ω . If $V = 50 \angle 0^\circ \text{V}$, $Z_1 = Z_2 = 10 + j24 \Omega$, find the value of Z_L for maximum power transfer. Also determine the maximum power. [2+3]

```
cx=-8i;% cx is the capacitive reactance
r=abs(5+12i+cx) % 2 marks
a=10+24i;b=r+cx;
z=a+((a*b)/(a+b));
it=50/z;
ir=abs(it*a/(a+b));
p=ir^2*r % 3 marks
```



6.4031 Ω

27.4048 W

2. Obtain $X(t)$ by solving the following state equations. Assume the initial conditions of the state variables are zero. The input is a unit step function. [6]

$$\dot{x}_1 = x_1 - 2x_2 + u$$

$$\dot{x}_2 = 3x_1 - 4x_2 + 6u$$

Check: Find the equilibrium point

```
A=[1 -2;3 -4];
b=[-1;-6];
x=inv(A)*b
```

$$x_1 = -4 + 9e^{-t} - 5e^{-2t}$$

$$x_2 = -1.5 + 9e^{-t} - 7.5e^{-2t}$$

3. The voltage transfer function of a linear electrical circuit is $G(s) = \frac{2}{s^2 + 2s + 8}$. If the steady-state response of the circuit is $2 + 4 \sin 3t$ Volts, determine the input voltage. [6]

$$\mathbf{8 + 12.1655 \sin(3t + 99.46^\circ) \text{ V}}$$

```
g=tf(2,[1 2 8]);
[mag,phase]=bode(g,3);
A=4/mag
Theta=-phase
```